

Physical Chemistry (I)

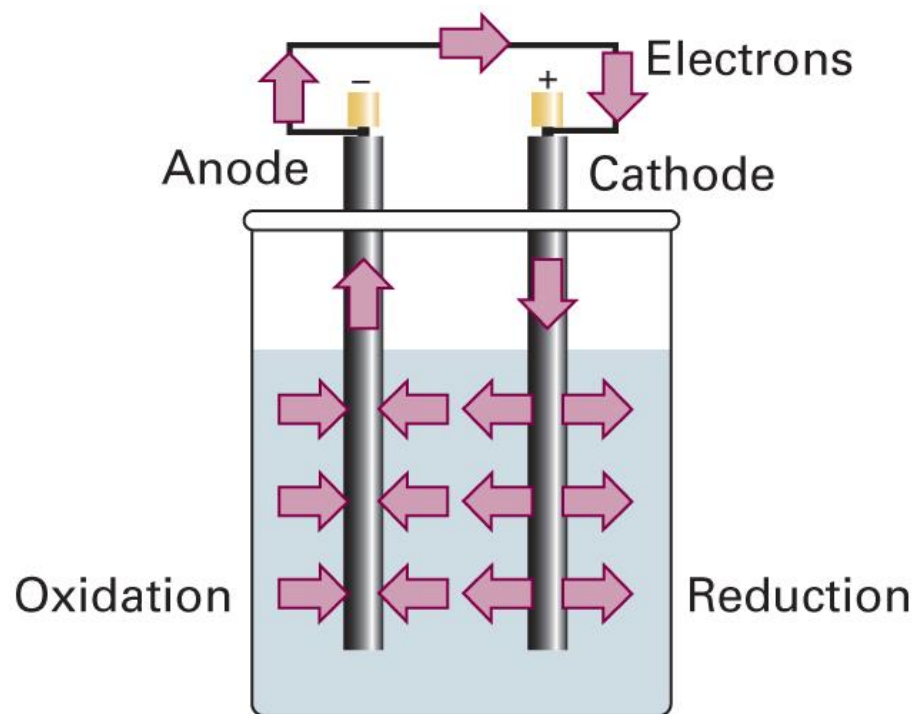
Lecture 22



In the Last Class

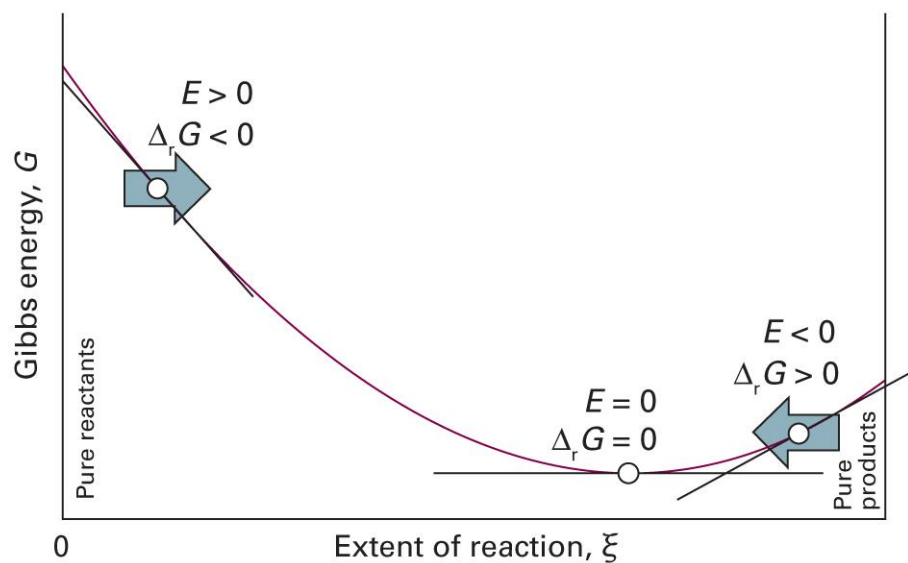
- Electrochemical cells
 - Basic terminology and concepts
 - Thermodynamics of cells
 - Electrode potentials
- Physical chemistry of ions
 - Gibbs energy of solvation: Born equation
 - Screening effects: Debye-Hückel equation

Electrochemical Cell



Thermodynamics of Cell

$$\Delta_r G = -\nu F E_{\text{cell}}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6C.4)

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

where $E_{\text{cell}}^{\ominus} = -\Delta_r G^{\ominus} / \nu F$

At equilibrium,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Gibbs Energy, Entropy, Enthalpy

$$E_{\text{cell}}^{\ominus} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G / \partial T)_p = -S$,

$$\frac{dE_{\text{cell}}^{\ominus}}{dT} = -\frac{1}{\nu F} \left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T} \right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F \left(E_{\text{cell}}^{\ominus} - T \frac{dE_{\text{cell}}^{\ominus}}{dT} \right)$$

Gibbs Energy of Solvation

From work for charging an ion,

$$\Delta_{\text{solv}} G^\ominus = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{1}{\epsilon_r}\right)$$

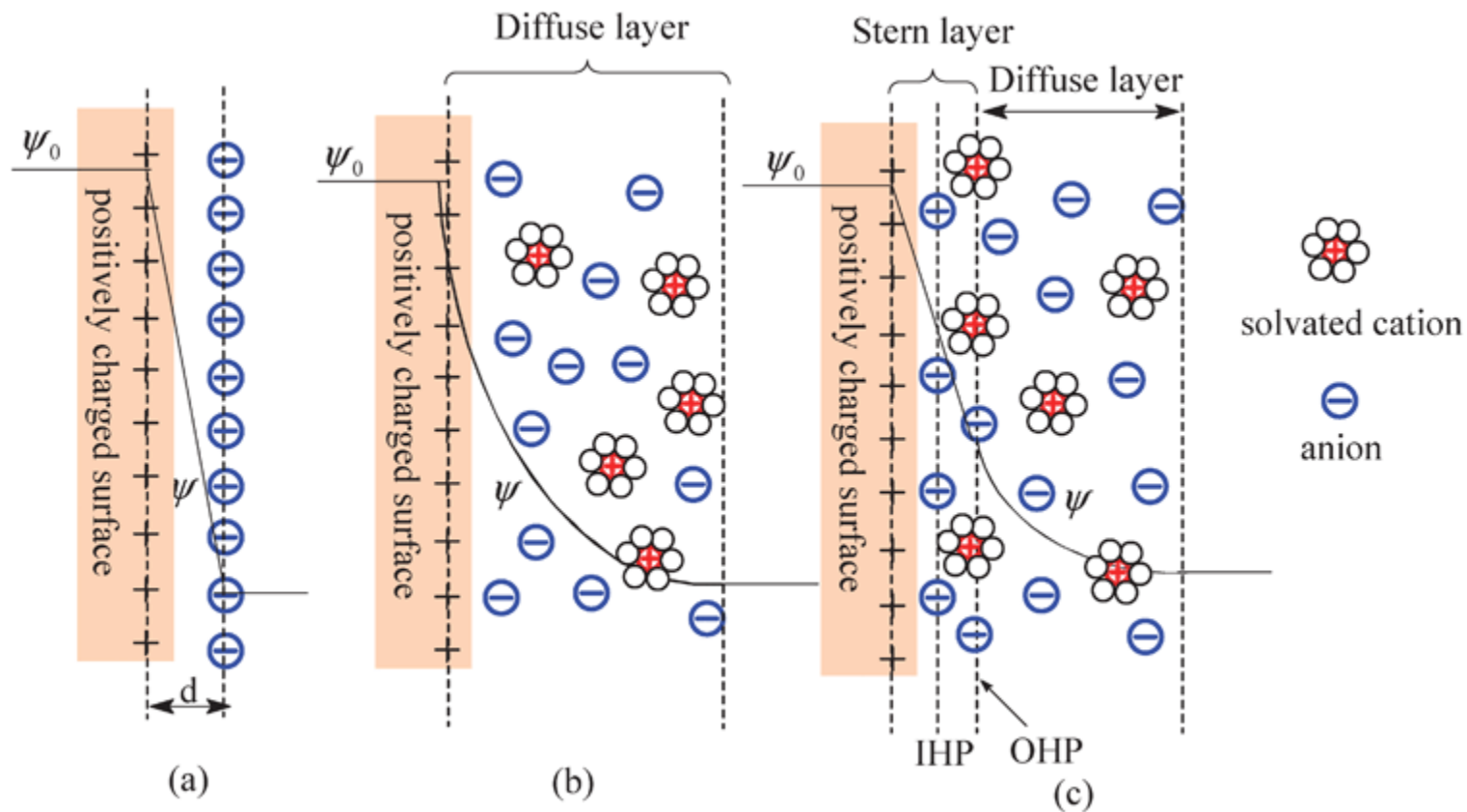
where the relative permittivity $\epsilon_r = \epsilon/\epsilon_0$.

Debye-Hückel Equation

- Boltzmann distribution of ions around a charge
- Poisson equation (charge density $\rightarrow$ electric potential)
- Poisson-Boltzmann equation
 - Differential equation for electric potential
 - Too complicated
- Dilute limit: Debye-Hückel equation
 - Using ion strength and Debye length, we obtain

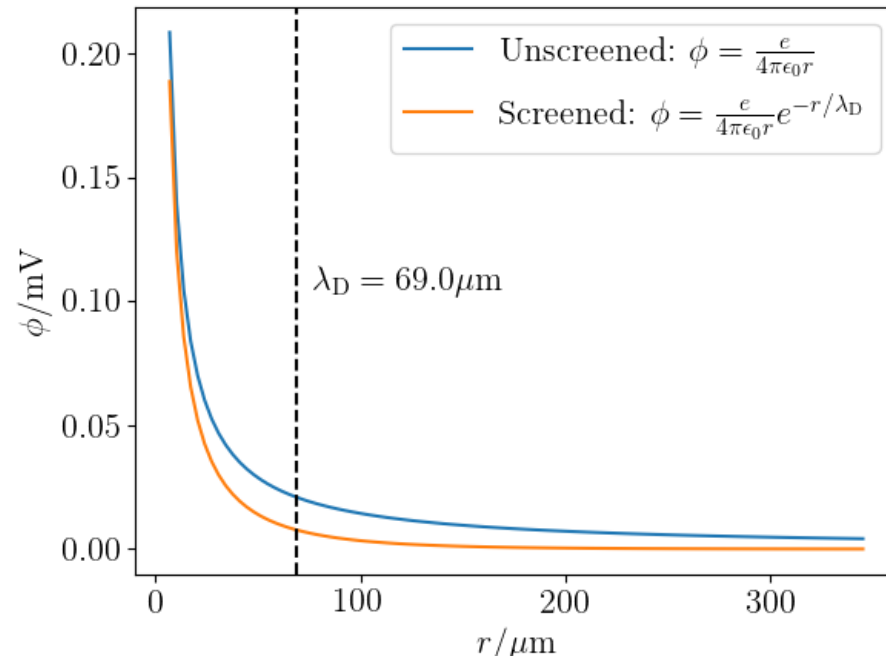
$$\nabla^2 \phi = \phi / r_D^2$$

Ions Around Electrodes



(Image: Zhang and Zhao, *Chemical Society Reviews* 38: 2520-2531 (2009), Figure 2)

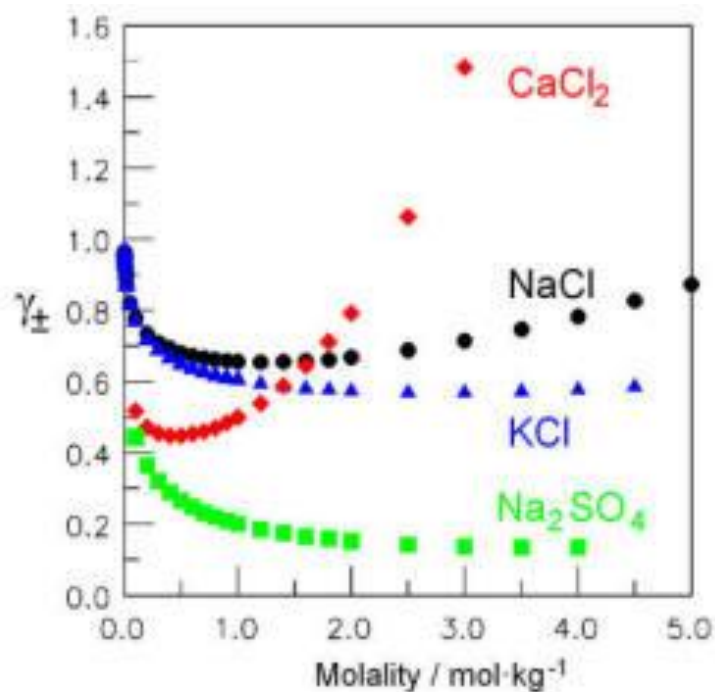
Ions Around Point Charge



(Image: <https://scipython.com/blog/the-debye-length>)

Activities of Ions

Quickly deviate from ideal behavior



(Image: <https://chem.libretexts.org/@go/page/44345>)

Molar Gibbs Energy of Ions

For an ideal solution consisting of M^+ and X^- ,

$$G_m^{\text{ideal}} = \mu_+^{\text{ideal}} + \mu_-^{\text{ideal}}$$

For a real solution,

$$\mu_J = \mu_J^\ominus + RT \ln a_J = \mu_J^\ominus + RT \ln(\gamma_J x_J) = \mu_J^{\text{ideal}} + RT \ln \gamma_J$$

and

$$G_m = \mu_+ + \mu_- = \mu_+^{\text{ideal}} + \mu_-^{\text{ideal}} + RT \ln \gamma_+ + RT \ln \gamma_-$$

Hence,

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+ \gamma_-)$$

Mean Activity Coefficients

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+ \gamma_-)$$

Experimentally impossible to separate γ_+ and γ_-

- **Mean activity coefficient:** geometric mean of γ_+ and γ_-

$$\gamma_{\pm} = (\gamma_+ \gamma_-)^{1/2}$$

Hence,

$$\mu_+ = \mu_+^{\text{ideal}} + RT \ln \gamma_{\pm}$$

$$\mu_- = \mu_-^{\text{ideal}} + RT \ln \gamma_{\pm}$$

Generalization

For a salt M_pX_q ($= pM^+ + qX^-$),

$$G_m = p\mu_+ + q\mu_- = G_m^{\text{ideal}} + pRT \ln \gamma_+ + qRT \ln \gamma_-$$

The mean activity coefficient is defined as

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/(p+q)}$$

so that

$$\mu_J = \mu_J^{\text{ideal}} + RT \ln \gamma_{\pm}$$

Activity and Work

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/(p+q)}$$

Since $G_m = G_m^{\text{ideal}} + pRT \ln \gamma_+ + qRT \ln \gamma_-$,

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+^p \gamma_-^q) = G_m^{\text{ideal}} + (p + q)RT \ln \gamma_{\pm}$$

Hence,

$$\ln \gamma_{\pm} = \frac{G_m - G_m^{\text{ideal}}}{(p + q)RT} = \frac{\overset{\text{work of charging an ion}}{w_e}}{(p + q)RT}$$

Work of Charging an Ion

- The work of charging an ion:

$$dw_e = \phi_{\text{others}} dQ$$

- The potential due to other ions:

$$\phi_{\text{others}} = \phi_{\text{DH}} - \phi_{\text{self}} = \frac{Q}{4\pi\epsilon r} e^{-r/r_D} - \frac{Q}{4\pi\epsilon r}$$

As we are interested in the vicinity of the ion ($r \rightarrow 0$),

$$e^{-r/r_D} \approx 1 - r/r_D$$

$$dw_e = -\frac{Q}{4\pi\epsilon r_D} dQ$$

Work of Charging an Ion

$$dw_e = -\frac{Q}{4\pi\epsilon r_D} dQ$$

For an ion of charge $z_i e$,

$$w_e = -\frac{1}{4\pi\epsilon r_D} \int_0^{z_i e} Q dQ = -\frac{z_i^2 e^2}{8\pi\epsilon r_D}$$

The molar work is

$$w_{e,m} = -\frac{N_A z_i^2 e^2}{8\pi\epsilon r_D} = -\frac{z_i^2 F^2}{8\pi N_A \epsilon r_D}$$

Total Work

$$w_{e,m} = -\frac{z_i^2 F^2}{8\pi N_A \epsilon r_D}$$

The total work of charging p moles of M^+ and q moles of X^- :

$$w_{e,\text{total}} = pw_{e,m}(M^+) + qw_{e,m}(X^-) = -\frac{(pz_+^2 + qz_-^2)F^2}{8\pi N_A \epsilon r_D}$$

Since $pz_+ + qz_- = 0$ (charge neutrality),

$$pz_+^2 + qz_-^2 = z_+(-qz_-) + z_-(-pz_+) = -(p+q)z_+z_-$$

$$w_{e,\text{total}} = \frac{(p+q)z_+z_-F^2}{8\pi N_A \epsilon r_D} = -\frac{(p+q)|z_+z_-|F^2}{8\pi N_A \epsilon r_D}$$

Activity

$$w_{\text{e,total}} = - \frac{(p + q)|z_+ z_-| F^2}{8\pi N_A \epsilon r_D}$$

$$\ln \gamma_{\pm} = \frac{w_{\text{e}}}{(p + q)RT} = - \frac{|z_+ z_-| F^2}{8\pi N_A RT \epsilon r_D}$$

Since $r_D = \{\epsilon RT / 2F^2 I b^{\ominus}(m_A/V)\}^{1/2}$,

$$\ln \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A} \left\{ \frac{b^{\ominus}(m_A/V)}{2\epsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

Debye-Hückel Limiting Law

$$\ln \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

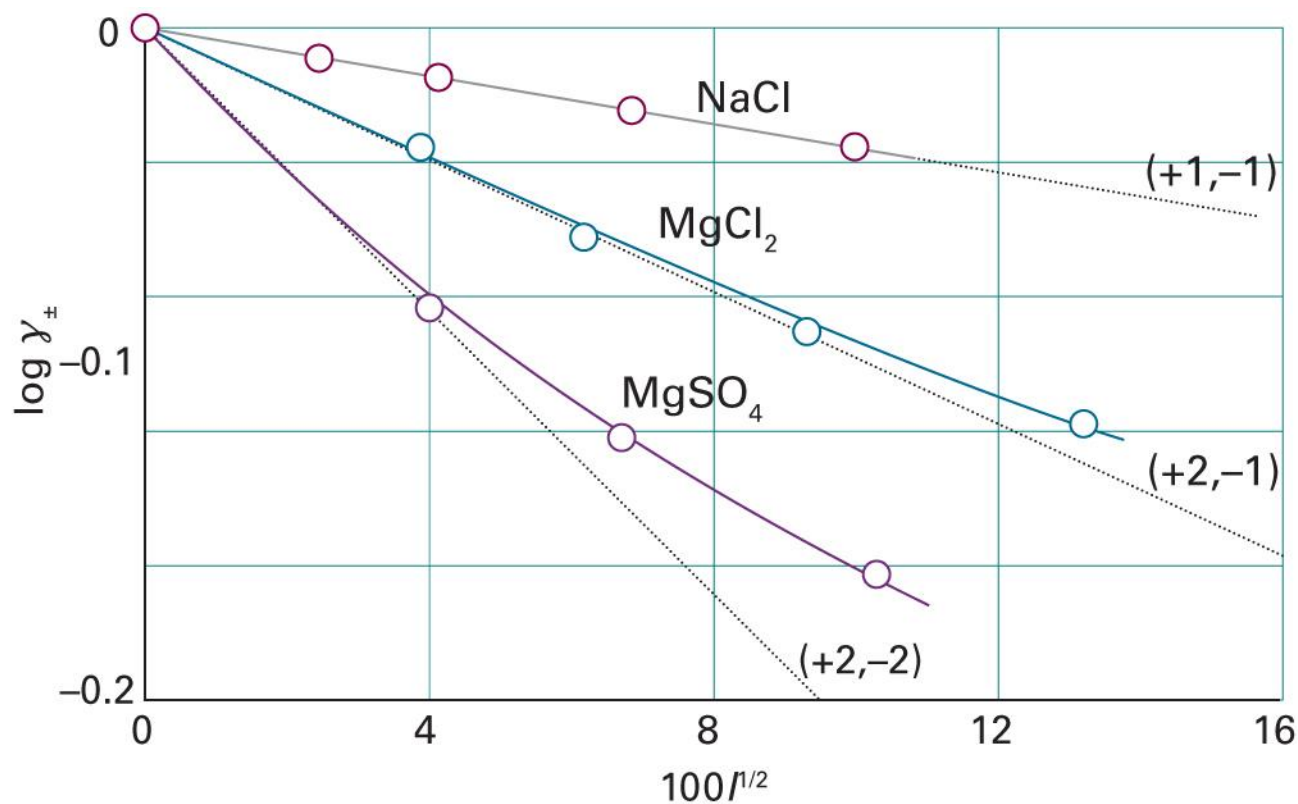
$$\log_{10} \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A \ln 10} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

Hence,

$$\log_{10} \gamma_{\pm} = -A|z_+ z_-| I^{1/2}$$

where $A = 0.509$ for an aqueous solution at 25°C

Debye-Hückel Limiting Law



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.5)

Brief Illustration 5F.4

Calculate the mean activity coefficient of 5.0 mmol kg⁻¹ KCl(aq) at 25°C.

The ionic strength is

$$I = \frac{1}{2} \sum_i \frac{z_i^2 b_i}{b^\ominus} = \frac{1}{2} \frac{b_+ + b_-}{b^\ominus} = \frac{b}{b^\ominus}$$

where $b = b_+ = b_-$. Then,

$$\log_{10} \gamma_{\pm} = -A |z_+ z_-| I^{1/2} = -(0.509) \times |(+1)(-1)| \times (5.0 \times 10^{-3})^{1/2} = -0.03$$

$$\gamma_{\pm} = 0.92$$

Extensions

- Debye-Hückel law (1923)

$$\log_{10} \gamma_{\pm} = -A|z_+z_-|I^{1/2}$$

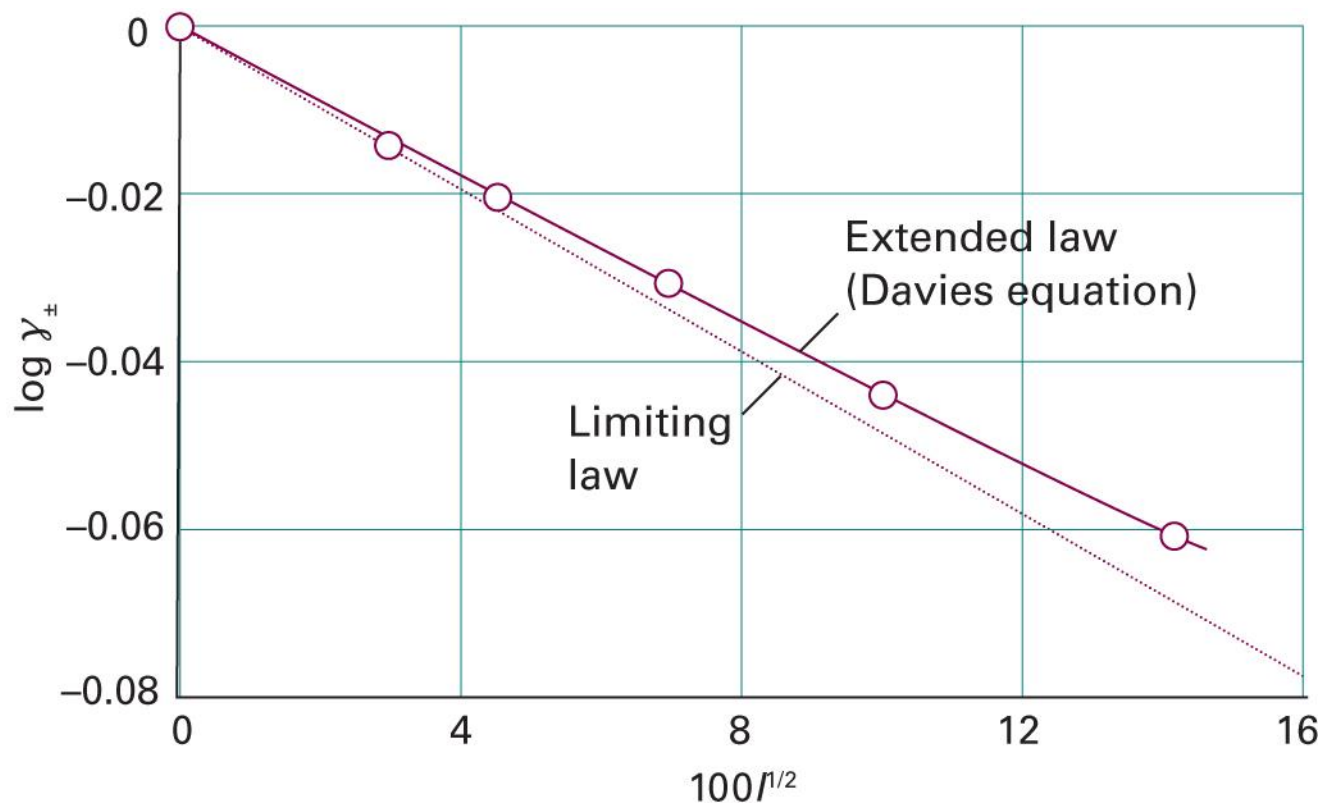
- Extended Debye-Hückel law

$$\log_{10} \gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1 + BI^{1/2}}$$

- Davies equation (1938)

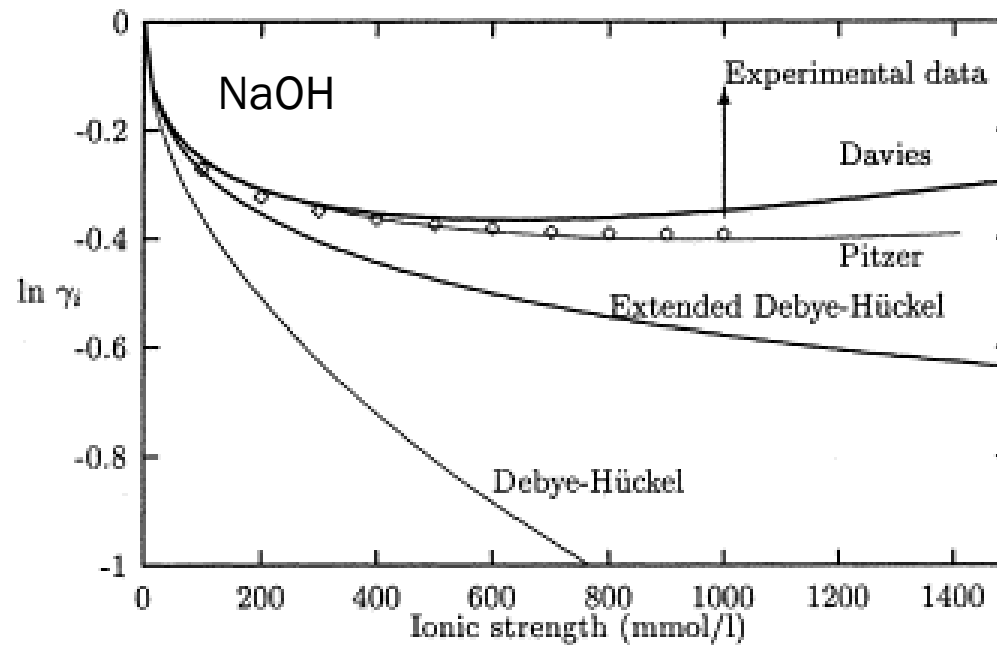
$$\log_{10} \gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1 + BI^{1/2}} + CI$$

Model Performances



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.6)

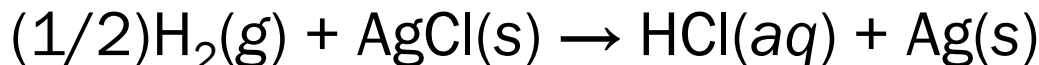
At Higher Concentrations



(Image: Samson et al., *Computational Materials Science* 15: 285-294 (1999), Figure 1)

Ion Activity and Cell Potential

Consider the Harned cell:



- Cell potential

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{AgCl/Ag, Cl}^{-}) - E^{\ominus}(\text{SHE}) = E^{\ominus}(\text{AgCl/Ag, Cl}^{-})$$

- Nernst equation

$$E_{\text{cell}} = E^{\ominus}(\text{AgCl/Ag, Cl}^{-}) - \frac{RT}{F} \ln \frac{a(\text{H}^{+})a(\text{Cl}^{-})}{\{a(\text{H}_2)\}^{1/2}}$$

Activities

$$E_{\text{cell}} = E^{\ominus}(\text{AgCl}/\text{Ag}, \text{Cl}^{-}) - \frac{RT}{F} \ln \frac{a(\text{H}^{+})a(\text{Cl}^{-})}{\{a(\text{H}_2)\}^{1/2}}$$

- Let $E^{\ominus} = E^{\ominus}(\text{AgCl}/\text{Ag}, \text{Cl}^{-})$
- At the standard pressure of 1 bar, $a(\text{H}_2) = 1$
- For the molality b of $\text{HCl}(\text{aq})$,

$$a(\text{H}^{+}) = \gamma_{\pm} b / b^{\ominus} \text{ and } a(\text{Cl}^{-}) = \gamma_{\pm} b / b^{\ominus}$$

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F} \ln \left(\frac{\gamma_{\pm} b}{b^{\ominus}} \right)^2$$

Molality Dependence

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F} \ln \left(\frac{\gamma_{\pm} b}{b^{\ominus}} \right)^2 = E^{\ominus} - \frac{2RT}{F} \left(\ln \gamma_{\pm} + \ln \frac{b}{b^{\ominus}} \right)$$

From the Debye-Hückel limiting law,

$$\begin{aligned} \ln \gamma_{\pm} &= (\ln 10) \log \gamma_{\pm} = -(\ln 10) A |z_+ z_-| I^{1/2} \\ &= -(A \ln 10) (b/b^{\ominus})^{1/2} \end{aligned}$$

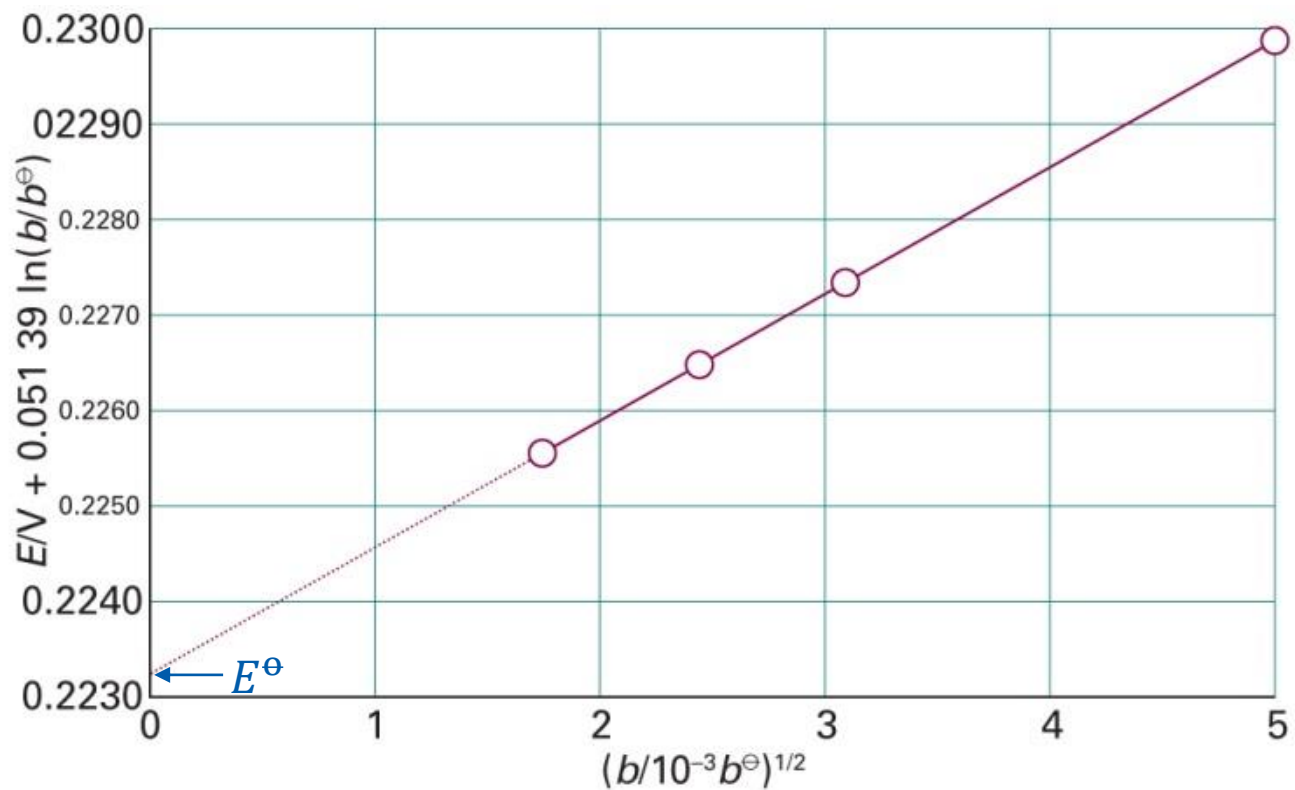
$$E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm} = E^{\ominus} + \frac{2A R T \ln 10}{F} \sqrt{\frac{b}{b^{\ominus}}}$$

Determination of $E^\ominus$

$$\underbrace{E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^\ominus}}_y = E^\ominus + \underbrace{\frac{2ART \ln 10}{F}}_a \underbrace{\sqrt{\frac{b}{b^\ominus}}}_x$$

From the plot of x versus y ,
we can obtain $E^\ominus$ as the y -intercept

Example



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 6D.1)

Determination of $\gamma_{\pm}$

$$E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm}$$

From the measurement of E_{cell} , we can obtain

$$\ln \gamma_{\pm} = \frac{F}{2RT} (E^{\ominus} - E_{\text{cell}}) - \ln \frac{b}{b^{\ominus}}$$

Brief Illustration 6D.3

The Harned cell has $E_{\text{cell}} = 0.46860 \text{ V}$ when $b/b^\ominus = 9.138 \times 10^{-3}$ at 25°C . What is the mean activity coefficient at this molality?

$$\ln \gamma_{\pm} = \frac{F}{2RT} (E^\ominus - E_{\text{cell}}) - \ln \frac{b}{b^\ominus}$$

$E^\ominus = 0.2232 \text{ V}$ for the Harned cell, and

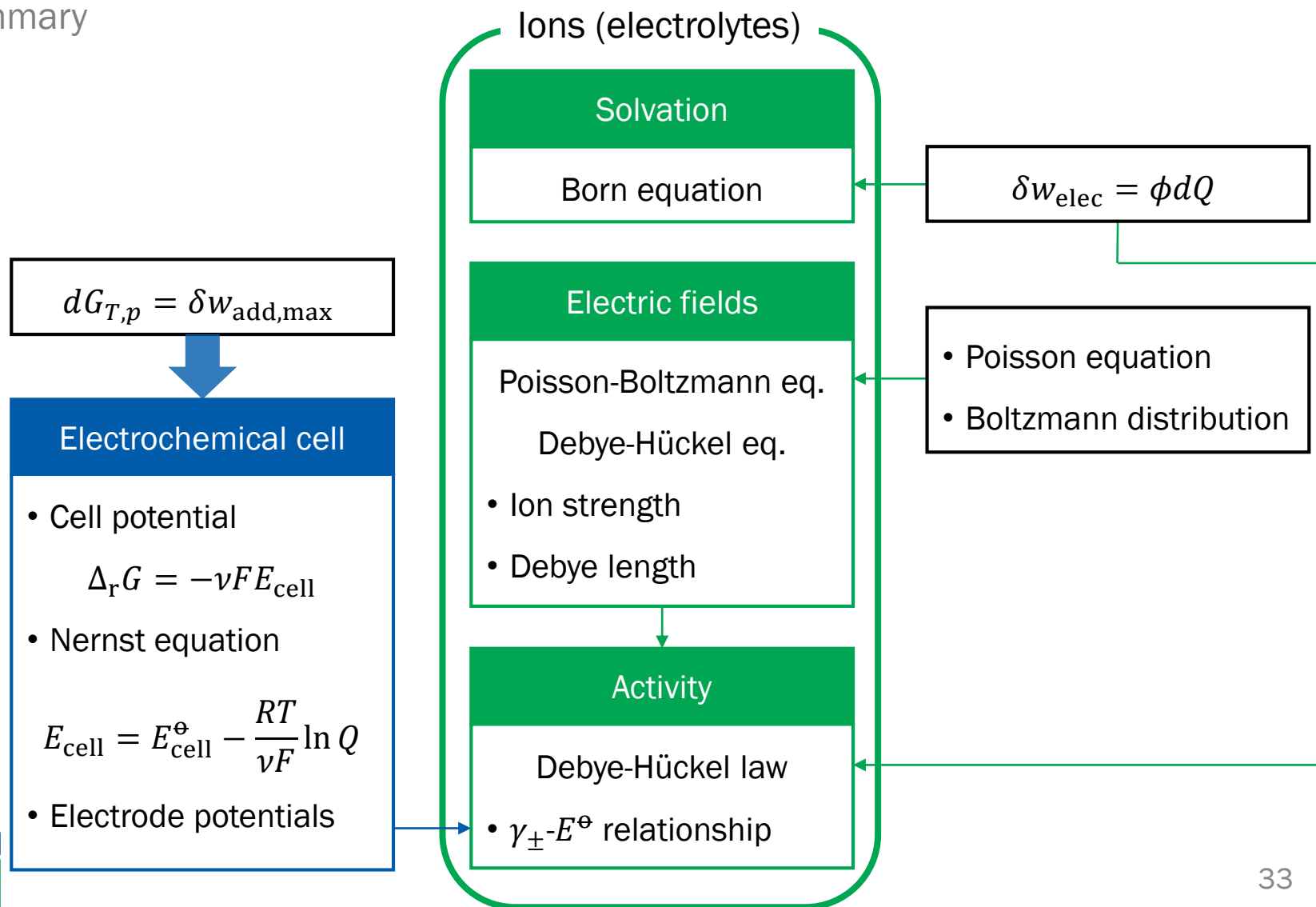
$$\frac{F}{2RT} = \frac{9.6485 \times 10^4 \text{ C mol}^{-1}}{2 \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})} = 19.46 \text{ V}^{-1}$$

Hence,

$$\ln \gamma_{\pm} = (19.46 \text{ V}^{-1}) \times (0.2232 \text{ V} - 0.46860 \text{ V}) - \ln (9.138 \times 10^{-3}) = -0.08$$

$$\gamma_{\pm} = 0.92$$

Summary



10. Statistical Mechanics

Chapter Overview

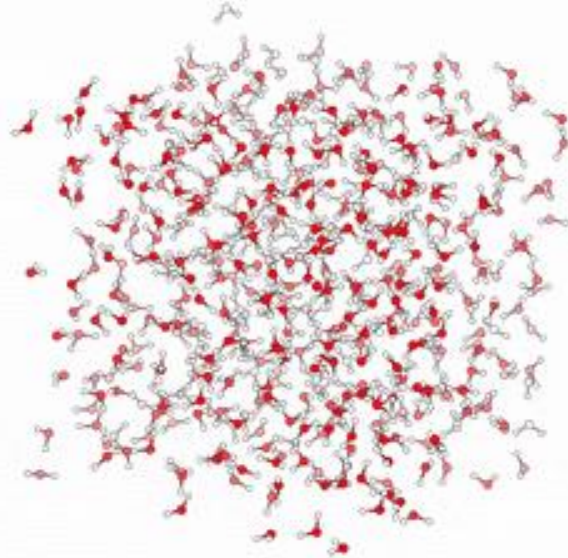
- Maxwell-Boltzmann statistics
 - Boltzmann factor and partition function
 - Classical statistical mechanics: phase space
- Partition function and thermodynamics
 - Generalization of partition function
 - Partition function and thermodynamic quantities
- Applications
 - Perfect gas and real gas
 - Equipartition theorem



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Statistical Mechanics

Thermodynamics (p , V , T , ...)



“Statistics” of
individual particles

(Image: https://commons.wikimedia.org/wiki/File:MD_water.gif)

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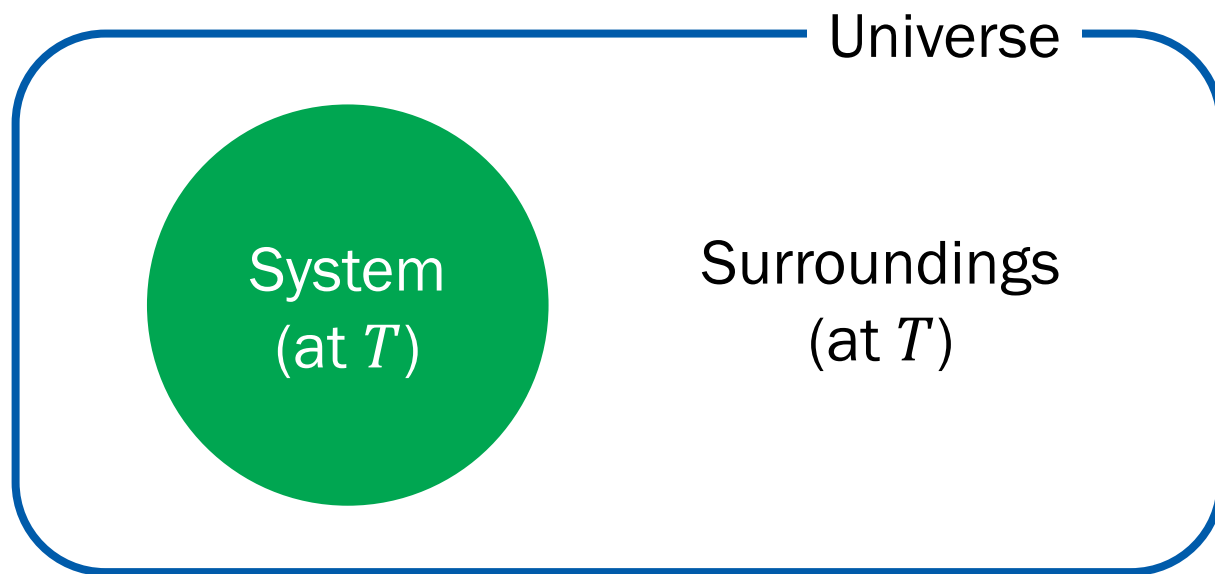
Maxwell-Boltzmann Statistics

Average distribution of independent particles
of a system in **thermal equilibrium**

Described by the **Boltzmann factor**

System Microstates

For a closed system with constant V and n ,



Consider two microstates s_1 and s_2 .

Probabilities

- Probability of having s_i is proportional to the number of microstates in the surroundings

$$P(s_i) \propto W_{\text{surr}}(s_i)$$

- Ratio between probabilities of having s_1 and s_2

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

Entropy Comes in

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

Using the statistical definition of entropy, $S = k_B \ln W$,

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{S_{\text{surr}}(s_2)/k_B}}{e^{S_{\text{surr}}(s_1)/k_B}} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_B}$$

What is $S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)$?

Entropy of Surroundings

Recall the fundamental equation of thermodynamics:

$$dU = TdS - pdV + \mu dn = TdS \text{ (at constant } V \text{ and } n)$$

For constant T , $\Delta U = T\Delta S$ and

$$\begin{aligned} S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1) &= \frac{1}{T} \{U_{\text{surr}}(s_2) - U_{\text{surr}}(s_1)\} \\ &= -\frac{1}{T} \{U(s_2) - U(s_1)\} \end{aligned}$$

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Written in System Variables

$$\frac{P(s_2)}{P(s_1)} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_B}$$

$$S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1) = -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

Combining the two equations,

$$\frac{P(s_2)}{P(s_1)} = e^{-\{U(s_2) - U(s_1)\}/k_B T} = \frac{e^{-U(s_2)/k_B T}}{e^{-U(s_1)/k_B T}}$$

Boltzmann Factor

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{-U(s_2)/k_B T}}{e^{-U(s_1)/k_B T}}$$

In general, for a microstate s with energy E ,

$$P(s) \propto e^{-E(s)/k_B T} \text{ (Boltzmann factor)}$$

or, using a constant q ,

$$P(s) = \frac{1}{q} e^{-E(s)/k_B T}$$

Determination of q

$$P(s) = \frac{1}{q} e^{-E(s)/k_B T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{q} e^{-E(s)/k_B T} = \frac{1}{q} \sum_s e^{-E(s)/k_B T}$$

$$q = \sum_s e^{-E(s)/k_B T}$$

(Partition function)

Brief Illustration 12A.2

Suppose that two conformations of a molecule differ in energy by 5.0 kJ mol⁻¹ (corresponding to 8.3×10^{-21} J for a single molecule), so conformation A lies at energy 0 and conformation B lies at $\varepsilon = 8.3 \times 10^{-21}$ J. What is the proportion of molecules in conformation B at 293 K?

$$E(A)/k_B T = 0 / \{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (293 \text{ K})\} = 0$$

$$E(B)/k_B T = (8.3 \times 10^{-21} \text{ J}) / \{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (293 \text{ K})\} = 2.05$$

The partition function is

$$q = \sum_s e^{-E(s)/k_B T} = e^{-E(A)/k_B T} + e^{-E(B)/k_B T} = e^0 + e^{-2.05} = 1.13$$

Brief Illustration 12A.2

$$q = 1.13$$

Hence, the proportion of molecules in conformation B is

$$P(\text{B}) = \frac{1}{q} e^{-E(\text{B})/k_{\text{B}}T} = \frac{1}{1.13} e^{-2.05} = 0.11$$

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Classical Statistical Mechanics

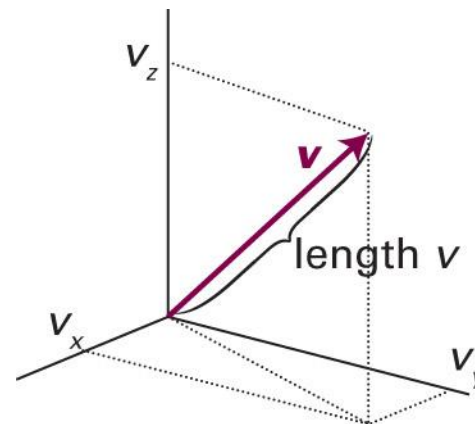
What is a **microstate** of a classical particle?

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$



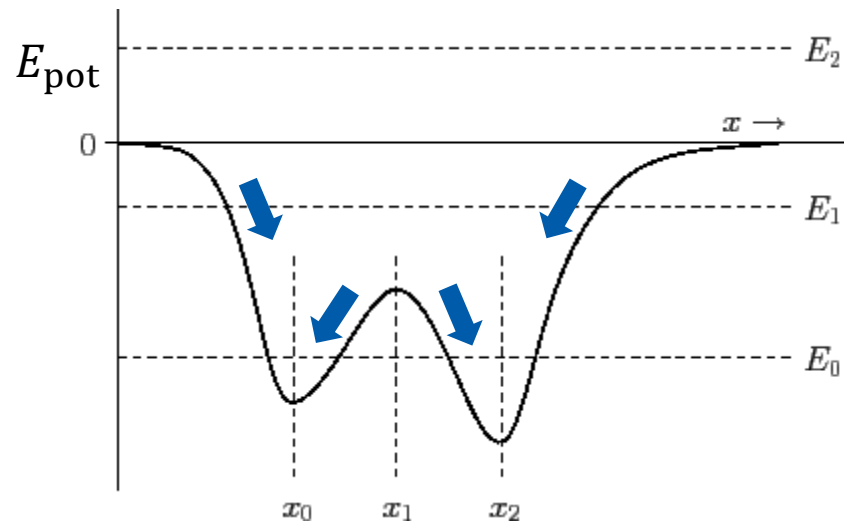
With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy ($-\nabla E_{\text{pot}}$)



(Image: <http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html>)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$-\nabla E_{\text{pot}} = m \frac{d^2\vec{r}}{dt^2}$$

↑
given

Second-Order ODE

$$\frac{d^2 \vec{r}}{dt^2} = - \frac{\nabla E_{\text{pot}}}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

Phase Space

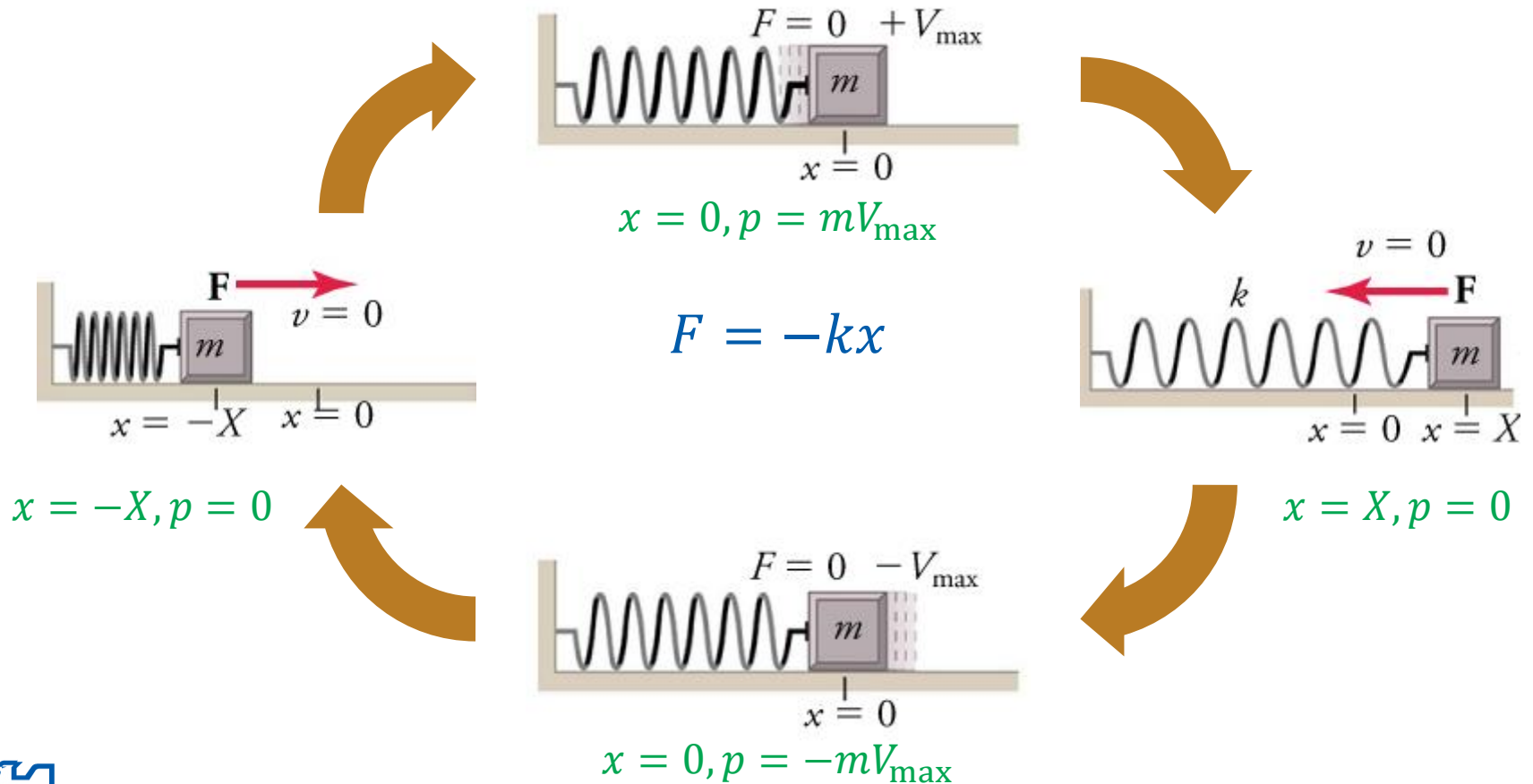
Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

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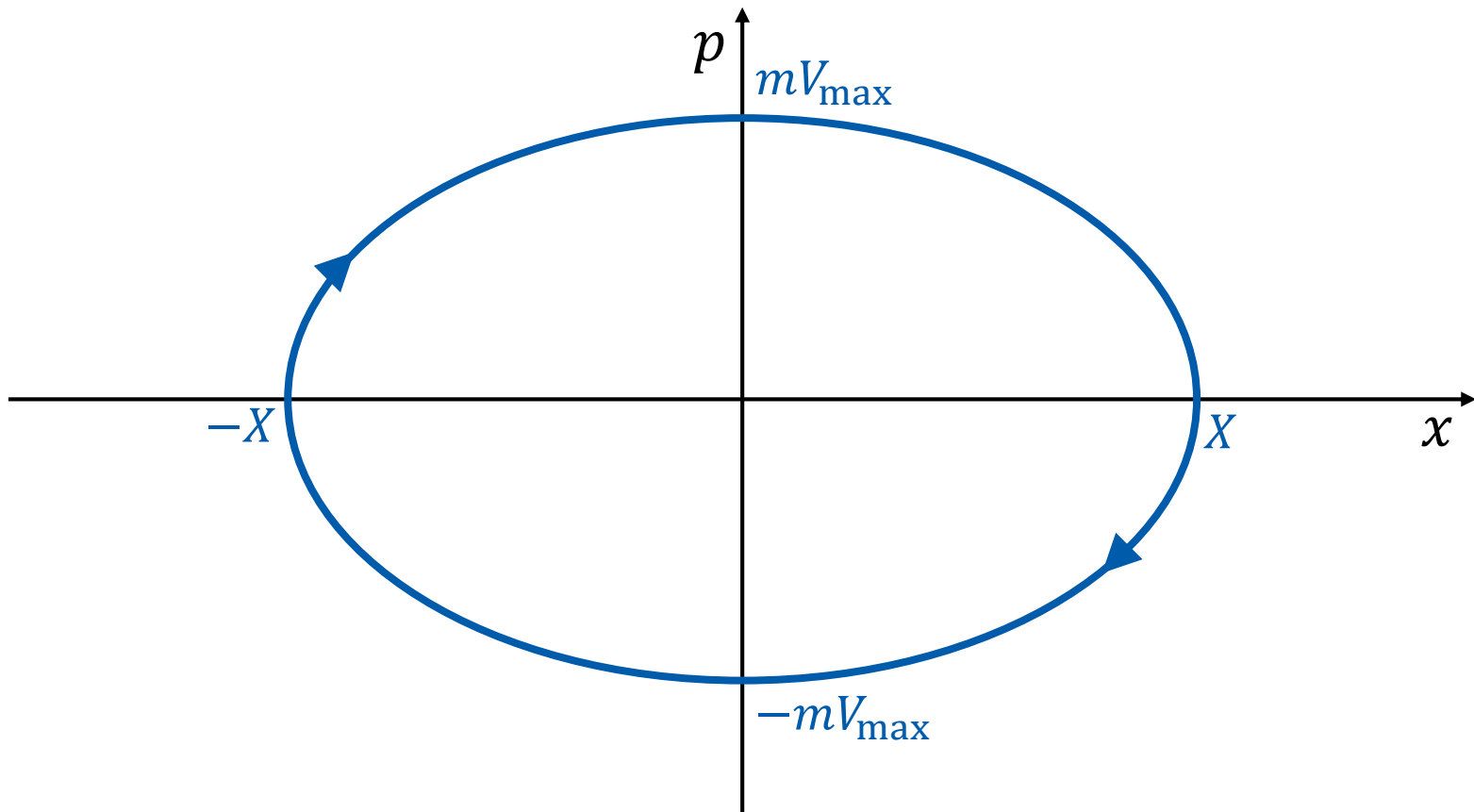
1-Dimensional Spring



(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

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Phase Space Representation



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Microstate of Classical Particle

